

Evidence-First VLM Cascades for Embedded Work-Zone State Estimation: Hyper-Parameter Portability and Counted Temporal Filters

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Abstract—We ask how closely a single work-zone-fine-tuned vision–language model (VLM), quantized to INT4 and running entirely on embedded automotive hardware, can approach such a pipeline while estimating four traversal states (OUTSIDE, APPROACHING, INSIDE, EXITING). Our *evidence-first cascade* operates one 2B VLM through asymmetric-cost prompt channels and hysteresis, under a *latency-honest* protocol where each system’s measured latency determines frame dropping. On 208 held-out ROADWork videos it matches a ported detector+CLIP baseline on accuracy and event recall but trails on precision, nuisance rate, and compute. Applying the 2B model’s temporal thresholds to an INT4 4B world model yields macro-F1 0.290 and INSIDE IoU 0.007; recalibrating the same four integers moves them to 0.515 and 0.574, so cross-model comparisons that inherit smoothing constants confound models with hyper-parameters. A joint estimator over detector and world-model evidence, with every parameter counted rather than set, reaches 0.546 at 25.9 false alarms per hour, inside the detector’s alarm budget, and keeps most of its advantage when fitted on one city and tested on the other; an ablation shows that the gain comes mainly from the temporal model rather than the fusion. These results establish computational viability, not safety readiness.

I. INTRODUCTION

Work zones concentrate difficult driving-perception conditions: their appearance is regional and unstandardized, geometry changes daily, evidence such as cones, barrels, signs, and workers is small or occluded, and both misses and nuisance alerts are costly. They accounted for roughly 101,000 U.S. crashes and 899 fatalities in 2023 [1], yet production ADAS rarely model work zones explicitly and temporary speed limits are often absent from digital maps.

The strongest onboard approach combines a YOLO-style work-zone detector, CLIP verification [2], exponential smoothing, and hysteresis to estimate four traversal states [3], [4]. It is fast and effective after a zone is established, but reported IoU falls from 0.56/0.47 on OUTSIDE/INSIDE to 0.11/0.09 on APPROACHING/EXITING [3]. These boundaries are precisely where an advisory is most useful: an alert after the vehicle reaches the barrels has little anticipatory value.

Meanwhile, fine-tuned VLMs describe work-zone scenes well offline [5], and weight-only quantization plus embedded runtimes bring small VLMs to automotive hardware [6], [7]. We ask whether a single fine-tuned VLM, operated through

cooperating prompt channels, can estimate traversal state in real time and compete with the detector pipeline under identical conditions.

Under an identical embedded protocol, the detector-free VLM is competitive on state coverage but trails on precision, nuisance rate, and compute. Two follow-ups shift the result. First, temporal thresholds fitted to one checkpoint are not neutral: inherited across models, they move a per-state IoU by $82\times$. Second, a 4B world model we quantize and calibrate, but never fine-tune, leads accuracy and boundary localization while the detector leads nuisance and precision. A joint estimator over both evidence streams, with parameters counted rather than set, resolves this trade-off, and most of its gain comes from the counted temporal model rather than from the fusion.

Answering the original question requires more than swapping models. At ~ 10 ms per output token, a full description costs roughly 500 ms while a two-token answer costs 75 ms: prompt choice is an economic decision. Our cascade therefore lets evidence quality gate smoothing (Section III-B): a transcribed sign can bypass a vote while a binary answer cannot.

Our contributions are: (1) a real-time, fully embedded traversal-state estimator from one quantized VLM; (2) latency-honest evaluation in which measured cycle time drives frame dropping, plus offline per-model recalibration; (3) paired comparison with a carefully reimplemented detector+CLIP port on the same 208 videos, protocol, and device; (4) evidence that temporal hyper-parameters do not transfer between models, with both checkpoints recalibrated symmetrically; (5) a joint estimator with counted rather than hand-set parameters, the best system we measure inside the detector’s false-alarm budget, with an ablation showing where its gain comes from; and (6) a leave-one-city-out and response-bias analysis on human labels that locates the residual gap in ego-relevance.

II. RELATED WORK

a) *Work-zone perception.*: Early systems used structured environmental models with sensor fusion [8] or frame-level classification [9]. ROADWork [5] contains 4,375 videos and 9.6K annotated images across 18 U.S. cities, and shows that

open-vocabulary models improve sharply after in-domain fine-tuning (+32.2 AP for detection and +36.7 SPICE for description). The closest system [3] deploys a ROADWork-trained YOLO detector with CLIP verification, EMA smoothing, and hysteresis, and defines the traversal-state evaluation we adopt.

b) VLMs for driving.: Large VLMs support end-to-end planning [10], [11], open-loop description [12], and reasoning [13], mostly on datacenter GPUs. We instead estimate one safety-relevant scene state under an embedded latency budget and study prompt-level failures, such as affirmation bias and distance blindness, that larger models can mask.

c) World models for physical AI.: Cosmos models jointly learn language, vision, and action for embodied agents [13], [14], mostly for planning at datacenter or automotive-SoC scale [15]. Rather than planning, we quantize a world model for Jetson-class perception.

d) Efficient inference and cascades.: AWQ [6] and embedded runtimes [7] enable multi-Hz VLM inference on embedded GPUs. Unlike classical cost-aware cascades [16], our stages share one model: prompts and token budgets create the cost asymmetry, and evidence semantics (transcribed text versus a binary answer), rather than scalar confidence, drive escalation.

III. SYSTEM

A. Two embedded deployments

We deploy *two* models on the same Jetson AGX Orin, adapted in the two ways such a model admits: one by task fine-tuning, one by quantization and per-model calibration alone. Both run the identical prompt channels, cascade, protocol and splits, so every comparison in this paper varies the model and its adaptation, never the pipeline around it.

Fine-tuned 2B. A 2B-parameter VLM (Qwen3-VL architecture [17], initialized from a physical-AI reasoning checkpoint [13]), used for perception only. We fine-tune it through a ROADWork-derived curriculum [5]: work-zone VQA (~46K pairs), ego-state supervision, sign reading, a general-driving retention mix, and ~10K work-zone-free BDD100K hard negatives. Before the last stage, the existence gate answered “yes” on 95% of clean scenes; after it, 0% on external negative controls. Positive-heavy intermediate training had destroyed calibration already present in the base model, and negative examples restored it.

The language backbone is quantized to INT4 (W4A16 AWQ, group size 128), the vision tower stays FP16, and both are compiled to TensorRT engines on-device [7] (1.38 GB language + 0.83 GB vision, ~4.5 GB peak runtime). A persistent process serves single-image requests. Costs decompose as 44.6 ms for vision encoding at 720p or ~20 ms at 480px, ~0.16 ms/input token for prefill, and 10.4 ms/output token for decoding. Output length therefore dominates, followed by input resolution; 480px preserved probe answers while saving 45 ms/cycle.

Quantized 4B world model. Cosmos3-Edge (C3E) [14] is a 4B physical-AI world model pairing an autoregressive reasoner with a trajectory-generating diffusion transformer. We quantize

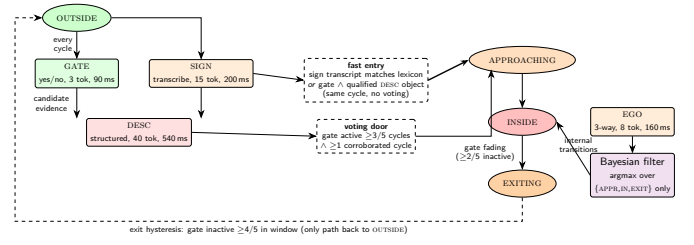


Fig. 1. Evidence-first cascade. In OUTSIDE, GATE and SIGN run every cycle and candidate evidence invokes DESC. Entry requires a sustained, corroborated vote or high-confidence same-cycle evidence. In active states, EGO drives a Bayesian filter; sustained gate clearance returns to OUTSIDE, while partial fading triggers EXITING. Constants are calibrated per model: 5/3/4/2 for the 2B and 3/3/2/1 for C3E, so the structure transfers but the numbers do not.

its reasoner to W4A16 AWQ INT4 (1.27 GB) and keep its visual tower in FP16 (983 MB), both as TensorRT engines on the same Orin; its frozen 26×46 patch grid fixes inputs at 736×416, and channel latencies are 162/182/488/556 ms (GATE/EGO/DESC/SIGN). It receives no work-zone training data and has never seen this benchmark. Building its engines required six patches to the embedded runtime; the patch set and deployment recipe are released with the code.

B. Prompt channels and their failure modes

We operate four fixed-prompt channels with fixed token budgets. Because the first generated token is an answer-start token, the minimum useful budget is two tokens.

a) Affirmation bias.: The GATE asks a neutral existence question and is calibrated on clean scenes, but binary prompts cannot be specialized by rephrasing. Variants asking whether work affects the vehicle’s path all answered “yes” on OUTSIDE probes, while distance prompts answered “close” regardless of scene. The learned discrimination resides in the trained question; prompt nuance does not transfer. Binary outputs are therefore weak evidence requiring a temporal vote, never sole triggers.

b) Faithful transcription.: The SIGN channel returns stable, verbatim text such as “SHOULDER WORK” and correctly reports absence. A binary rephrasing misses signs that transcription catches on the same frames. We therefore transcribe and derive a binary value by matching a work-zone keyword lexicon.

c) Structured corroboration.: The DESC channel’s object–location prefix often verbalizes disqualifying context—e.g., “barriers on right sidewalk” or “work vehicle parked.” Corroboration therefore requires a work-zone object without such a qualifier. Prompts, budgets and lexicons are released with the code.

C. Evidence-first cascade

The cascade (Fig. 1) allocates compute by state.

Outside (patrol). GATE and SIGN run every cycle (~315 ms, 3.2 Hz); either can invoke DESC. Entry into APPROACHING has two paths: a voting path, with the gate active in at least three of five cycles and one corroboration, or fast entry on a lexicon-matching sign or gate+DESC agreement in the same cycle. Fast

entry avoids the ~ 1.6 s voting delay for evidence such as a transcribed “ROAD WORK AHEAD.”

Active states. EGO updates a Bayesian filter whose argmax is restricted to the three active states. Otherwise, a single “outside” reply immediately after sign-triggered entry would erase the sign’s anticipatory value. Only sustained gate clearance returns the system to OUTSIDE; partial fading triggers EXITING, which EGO never produced even when offered explicitly. DESC refreshes every 1.5 s. All thresholds ($N=5$, $K_{\text{enter}}=3$, $K_{\text{exit}}=4$, $K_{\text{fade}}=2$) were fixed on calibration data.

D. Joint estimator over both evidence streams

Our first attempts to combine the detector and the world model grafted one onto the other’s temporal machinery, and all of them failed. The ported detector counts cycles (a 150-cycle maximum approach duration, 20 minimum frames outside) tuned for its 43 ms loop; the cascade votes over a window tuned for a ~ 700 ms stride. Reusing either as the spine imports constants that are wrong for the combination, which is the threshold-inheritance problem of Section VI one level up.

The joint estimator therefore inherits nothing: it has no EMA, no hand-set hysteresis, no threshold and no cycle counter. It is a forward filter over the four states. Its observation combines the detector’s calibrated frame score (six buckets), the world model’s gate, sign-keyword and corroboration answers ordered by evidence strength (six levels), and the age of that answer (three buckets), for 108 discrete cells. The emission $P(\text{observation} \mid \text{state})$ is counted directly on the calibration split with Laplace smoothing, and it is joint rather than factorized because the interaction is the signal: the world model’s APPROACHING is reliable when the detector corroborates it and mostly a false alarm when it does not (73% of those frames are OUTSIDE). The transition is a per-second rate matrix, counted as ground-truth transitions over dwell time and scaled by the elapsed dt of each step, so it does not depend on the loop rate. Finally, the log-likelihood is scaled by dt/τ with $\tau = 0.5$ s: a world-model answer persists across roughly 30 detector frames, and applying its likelihood once per frame treats 30 correlated looks as independent, which drove an early version to 135 false alarms per hour.

At run time the detector runs at its native rate and is never blocked. The world model is polled asynchronously and re-polled as soon as it answers; its last answer is held, with its age as part of the observation.

IV. EVALUATION PROTOCOL

a) Benchmark. We use the detector baseline’s 520-video ROADWork-derived traversal benchmark [3]: Boston and Seattle footage at 1920×1080 , 30 fps, with 484K interval-labeled frames. A state-stratified 60/40 video split assigns all design choices and thresholds to calibration (312 videos, 2.59 h) and final results to validation (208 videos; 192,135 frames, 1.78 h, 13 pure-negative videos); the benchmark totals 4.37 h of annotated driving. All systems see the same split. All 25 drives occur in both partitions, so they are not

TABLE I
VALIDATION SPLIT, 208 VIDEOS, ONE PROTOCOL, ONE DEVICE (EVENT P/R IN %, TIMING IN S). RAW: THE CASCADE OF SECTION III-C, SAMPLED DECODING; REC.: GREEDY PLUS A 1 S INSIDE DEBOUNCE; C3E: THE WORLD MODEL WITH ITS OWN THRESHOLDS; JOINT: THE ESTIMATOR OF SECTION III-D, FITTED ON 312 CALIBRATION VIDEOS.

Metric	Det.+CLIP	2B fine-tuned		C3E	Joint
	[3]	raw	rec.		
Frame acc.	62.9%	62.5%	62.0%	65.5%	66.1%
Macro F1	0.470	0.464	0.453	0.515	0.546
IoU OUT	0.617	0.586	0.581	0.615	0.641
IoU APPR	0.124	0.152	0.152	0.278	0.246
IoU IN	0.544	0.535	0.532	0.574	0.578
IoU EXIT	0.106	0.085	0.062	0.071	0.160
IN ev. recall	95.7	96.8	96.8	87.0	85.9
IN ev. prec.	79.9	54.1	76.7	71.5	66.8
False alarms/h	30.9	60.1	37.7	64.6	25.9
Entry MAE	2.52	2.53	2.52	2.15	2.22
IN entry med.	3.47	2.73	2.77	2.50	2.27

route-independent (Section IX). Interval labels are evaluation-only and hard negatives come from BDD100K. Fine-tuning images and benchmark videos both derive from ROADWork, so depicted-zone overlap is possible; the ROADWork-trained baseline shares this condition.

b) Latency-honest camera simulation. Embedded perception cannot pause the world. After each measured inference cycle of duration Δ on the target hardware, the replay advances by Δ and drops intervening frames exactly as a live camera would. Slower systems therefore observe fewer, sparser frames and pay latency inside the accuracy result rather than beside it. Predictions are held between cycles, matching a driver-facing display, and scored per frame.

c) Metrics. We adopt the baseline’s suite [3]: frame accuracy and macro-F1; per-state IoU; event precision/recall, where an interval is correct if it overlaps one of the same state; matched-event entry offset; and transition matching at $\pm\{0, 5, 15, 30\}$ frames. Videos beginning in a state contribute no entry event. Event metrics capture whether the warning occurs at all, whereas IoU and timing capture boundary quality; both are needed because a long, early interval can have perfect recall but poor localization. Confidence intervals and paired tests use 10,000 video-level bootstrap samples, resampling the same video indices for paired systems.

V. RESULTS: ONE FINE-TUNED MODEL VS. THE DETECTOR

A. Comparison with the deployed detector system

Table I places every system on the same videos, device and protocol. This section reads its first three columns; C3E is the outcome of Section VI and the joint estimator of Section VII. Against the ported detector+CLIP pair, the fine-tuned 2B is statistically tied on frame accuracy and INSIDE event recall, and shows a small APPROACHING-IoU advantage (0.152 vs. 0.124) whose paired 95% CI marginally includes zero ($[-0.005, +0.058]$, 10,000 video-level replications), but trails badly on event precision (54.1% vs. 79.9%). This is

a competitive detector-free result rather than a replacement result. Greedy decoding plus a 1 s `INSIDE` debounce recovers most of the precision gap (76.7%) without retraining, at a small cost in boundary IoU. Median entry error is comparable across systems (1.27 s for the debounced 2B against 1.37 s for the detector), while the 2B’s `INSIDE` entry median is better by 0.70 s.

The four-state pattern matters more than aggregate accuracy. Both learned pipelines perform well on established `INSIDE` intervals but poorly on `EXITING`; the VLM’s main advantage is `APPROACHING`, where text and distant semantic evidence appear before detector counts become stable. Conversely, the detector’s stronger `OUTSIDE` IoU and lower nuisance rate show why semantic anticipation cannot be accepted unconditionally. This state-wise complementarity, rather than a small difference in overall accuracy, is the result that survives.

B. Operational false-alarm characterization

A driver-facing system is judged by nuisance rate, not only event precision. Over 1.78 h, false active-state events occur at 60.1/h; greedy plus 1 s debounce cuts this to 37.7/h (2 s: 27.5/h), versus 30.9/h for the detector. Specificity is measured on 75,918 `OUTSIDE` frames (70.6% correctly held) and on all 36 pure-negative videos, where the system remains `OUTSIDE` for 67.4% and 58.0% of their frames respectively. Residual errors are predominantly the ego-relevance family of Section VIII, not clean-scene noise. Of 155 matched `INSIDE` entries, 81.9% fire early (median 2.4 s), which an advisory prefers but which stretches intervals.

C. Ablations and design validation

Seven single-toggle ablations show that no component moves frame accuracy or macro-F1 by more than 1.5 points, yet each protects a targeted failure. The sign channel must transcribe rather than classify: an equal-cost binary question is the most damaging change, reducing accuracy 62.5%→52.5% and `INSIDE` precision to 46.1%. Removing the channel barely moves aggregates because its value is anticipatory entry rather than frame correctness. Greedy decoding costs 0.5 accuracy points but improves precision at matched recall, making it our reproducible default. Fast entry, qualifier filtering, and active-state filtering protect behaviors poorly captured by aggregate metrics; the qualifier filter is largest, adding 2.2 points concentrated in pure-negative videos.

D. Operational cost and cross-city consistency

At 480px, p50/p95 latency is 83/90 ms (`GATE`), 136/149 ms (`EGO`), 155/169 ms (`SIGN`), and 468/508 ms (`DESC`). Patrol and active cycles run at p95 259 ms (3.9 Hz) and 239 ms (4.2 Hz); the heaviest path is 768 ms (1.3 Hz). Here “real time” means keeping pace with a live camera at these rates, not meeting a specific alert deadline. Under load, the VLM draws 36.5 W median (42.5 W peak; 7.9 W idle) and 3–26 J/cycle, versus ~24 W and ~1 J for the 43 ms detector: 1.5× the power and 3–26× the energy per decision. Each channel currently re-encodes the frame; sharing its visual embedding is an obvious route to reduce the gap.

VI. VARYING THE MODEL: CALIBRATION AND GENERALIZATION

Section V held the model fixed while varying its pipeline, confounding the checkpoint with thresholds fitted to it. The base-checkpoint ablation changes only the former and appears to settle the question, but inherits the latter. We separate the two factors using the second deployment of Section III-A, recalibrating each checkpoint on its own evidence. This changes two conclusions above and produces our strongest system.

a) Temporal hyper-parameters do not transfer.: With the 2B thresholds, C3E scores macro-F1 0.290 and `INSIDE` IoU 0.007 at 3.8% recall. Taken literally, this would mean that C3E cannot recognize being inside a zone. It is an artifact. Recalibrating the same four integers for C3E moves these to 0.515, 0.574, and 87.0% on validation: `INSIDE` IoU rises 0.567 absolute and recall 83.2 points, with model, prompts and videos fixed; the 82× ratio is large mainly because the inherited value is near zero. The search covers $N \in \{3, 5, 7, 9\}$, $K_* \in 1..N$ and the `EGO/SIGN` switches (4,896 configurations), scored by calibration macro-F1. The selected configuration ($N=3$, $K_{\text{enter}}=3$, $K_{\text{exit}}=2$, $K_{\text{fade}}=1$) is shorter and less hysteretic, consistent with less noisy per-frame evidence, and disables `EGO` (Section VIII).

We make per-model recalibration cheap by recording each channel’s output once per sampled frame and replaying the state machine, reducing a threshold sweep from GPU-hours per configuration to CPU-seconds.

Replay evaluates only temporal logic over a fixed record. The identical search on the identical videos barely moves the fine-tuned 2B (macro-F1 0.453→0.459, against 0.290→0.515 for C3E): a search that merely rewarded flexibility would have helped both models equally.

The same procedure also measures fine-tuning under matched conditions. Recalibrating the *base* checkpoint—identical family, scale, architecture and visual tower, differing only in the work-zone curriculum—lifts it from macro-F1 0.281 to 0.383 on calibration, against 0.451 to 0.496 for the fine-tuned version. Given their own thresholds, fine-tuning is worth +0.113 macro-F1 and +8.8 accuracy points, not the near-total collapse (`INSIDE` IoU 0.535→0.012) that the base checkpoint shows under the fine-tuned model’s thresholds. It buys a real and specific capability in domain. Thus the 82× change isolates fitting the wrapper to its model rather than extra visual training or validation leakage.

Notably, calibration disables the `EGO` channel entirely. A seemingly task-specific question can thus reduce end-to-end accuracy even for a physical-AI world model; the cascade performs better when direct evidence and temporal structure infer state.

Every decision in this paper is made on the 312-video calibration partition and none on validation: prompts, budgets and the qualifier lexicon on the full partition, and per-model thresholds by replaying recorded evidence over a 100-video subset of it. The 208 validation videos enter only as the reported result.

b) No configuration dominates.: The two deployments fail complementarily: C3E leads accuracy, macro-F1 and both

boundary IoUs, but at 64.6 false alarms/h against the detector’s 30.9; the detector leads nuisance and INSIDE precision but localizes boundaries poorly. As single models, neither is the right choice for every application. Section VII shows that combining their evidence under a temporal model fitted for the pair resolves this trade-off.

VII. COMBINING THE DETECTOR AND THE WORLD MODEL

The joint estimator of Section III-D reaches macro-F1 0.546 at 25.9 false alarms per hour (Table I). It is the only system we measure that improves on the detector’s quality while staying inside the detector’s alarm budget, and the only one that moves EXITING, which every single-model system leaves between 0.06 and 0.11. Against the detector (paired video-level bootstrap, 10,000 replications), the gains in accuracy, macro-F1 and the three non-OUTSIDE IoUs exclude zero, the false-alarm difference does not (-5.1 , $[-13.7, +3.8]$), and INSIDE event precision is 0.131 lower. Against the calibrated world model it gains 0.031 macro-F1 and 0.089 EXITING IoU and removes 38.8 alarms per hour.

We report the fit on all 312 calibration videos; a 100-video fit scores the same within noise but at 33.2 alarms per hour, violating our pre-set criterion (maximize macro-F1 with false alarms no worse than the detector’s 30.9).

a) *Operational profile.*: The detector drives the clock and is never blocked (p50 44 ms, p95 59 ms), so the state updates at 22.1 Hz. The world model answers asynchronously at 0.62 Hz (p50 1.54 s between answers). The estimator therefore runs at the detector’s rate while carrying world-model evidence, rather than at the slower model’s rate as our earlier chained designs did.

b) *Where the gain comes from.*: A channel ablation, which refits the filter without each stream, is uncomfortable for the fusion story. Removing the world model entirely costs 0.010 macro-F1 and *improves* the alarm rate (25.9 to 18.5 per hour) and INSIDE precision (66.8% to 76.5%). Removing the detector costs 0.086 macro-F1 and collapses EXITING IoU from 0.160 to 0.017. Of the world-model channels only DESC pays for itself in domain; removing GATE slightly helps. The detector alone inside the counted filter scores 0.536, against 0.470 for the same detector with its hand-tuned state machine. Most of the in-domain improvement is therefore the temporal model, not the fusion: counting the parameters is worth 0.066 on the detector’s own evidence, and adding the world model on top is worth 0.010. On this benchmark, the honest recommendation would be the detector inside the counted filter, at 18.5 false alarms per hour.

c) *A learned alternative.*: A strictly causal multi-stage TCN [18] trained on the same evidence ties the best macro-F1 (0.550) but at 71.9 false alarms per hour and 55.1% INSIDE precision, failing the same criterion.

d) *Across cities.*: Because all drives appear in both splits (Section IV), we also count the estimator’s tables on one city only and replay it on the other city’s validation videos (Table II). Fitted on Boston and tested on Seattle it loses 0.020 macro-F1; fitted on the 66 Seattle videos and tested

TABLE II
JOINT ESTIMATOR FITTED ON ONE CITY, REPLAYED ON THE OTHER CITY’S
VALIDATION VIDEOS. DETECTOR+CLIP: 0.462 (SEATTLE), 0.469 (BOSTON).

Test city	Fitted on	Macro F1	IoU IN	IN prec.	FA/h
Seattle (54)	all 312	0.519	0.620	61.3	38.1
	Boston only (246)	0.499	0.618	60.7	42.6
Boston (154)	all 312	0.551	0.560	68.9	21.8
	Seattle only (66)	0.513	0.421	73.1	32.3

TABLE III
GATE “YES” RATE BY TRUE STATE, INSIDE-VS-OUTSIDE AUC, AND DESC
CORROBORATION ON OUTSIDE; SAME 100 LABELED VIDEOS, 1 Hz.

Checkpoint	GATE “yes” when truth is				AUC	DESC
	OUT	APPR	IN	EXIT		
2B base	3.8%	53.1%	71.8%	9.8%	0.84	31.3%
2B fine-tuned	42.4%	85.3%	92.2%	51.2%	0.75	10.7%
C3E zero-shot	29.4%	93.3%	94.0%	34.4%	0.82	35.7%

on Boston it loses 0.038, mostly in INSIDE IoU. Both remain above the detector pipeline in the same city. The language models themselves were not re-fine-tuned per city, so this bounds memorization by the temporal model only.

VIII. FAILURE ANALYSIS: THE EGO-RELEVANCE GAP

Nearly all residual false-positive mass has one signature: descriptions are factually correct about objects irrelevant to the vehicle’s path. Examples include a parked work vehicle more than 60 m away, sidewalk reconstruction behind barriers, and residual detour signs for another street. The qualifier filter catches only cases where the model verbalizes this context; it does not emit qualifiers reliably, and binary or distance prompts do not recover the judgment.

a) *Response bias.*: The three checkpoints were recorded on the same labeled videos, which separates how often a model says “yes” from how well its answer tracks the state (Table III). The base 2B rarely answers yes outside a zone and its gate separates INSIDE from OUTSIDE best (AUC 0.84); fine-tuning moves the same gate to 42.4% and 0.75. OUTSIDE in ROADWork includes visible work that does not concern the vehicle, so a “yes” there is often correct perception and missing ego-relevance. The fine-tuned model’s advantage under matched calibration therefore does not come from its gate: its description corroborates on 10.7% of OUTSIDE seconds against 31.3% for the base model.

The natural reading is incomplete fine-tuning coverage: empty hard negatives do not teach that visible work-zone objects can be irrelevant. A control shows that this is not the whole explanation. C3E transcribes signs correctly and emits plausible 2D boxes, yet labels most verified INSIDE frames “off-road/sidewalk” in ego-lane probes, nearly independently of ground truth. It exhibits the same failure with the same sign as the fine-tuned model. Geometric prompting also fails: an ego corridor built from the model’s own boxes inherits the query’s affirmation bias, because “locate all work-zone

objects” presupposes their existence and produces boxes in 6/6 pure-negative probes. Geometry cannot correct bias in its own input.

Perceiving objects and judging whether they claim the ego path are distinct capabilities that prompting does not bridge across two model families, scales and adaptation strategies; the gap belongs to the interface, not one fine-tuning run. This also explains uniformly weak EXITING IoU (0.085 for 2B, 0.106 detector, 0.071 C3E): visible but receding objects require ego-relative reasoning that neither detection counts nor detection-flavored questions capture. The joint estimator reaches 0.160 on EXITING only through the detector: without it, EXITING IoU falls to 0.017. The detector’s object score decays as objects recede, and the counted filter reads that decay directly; no language channel supplies it.

Trajectory-grounded supervision is the natural remedy. ROADWork’s 129K path annotations [5], unused in our fine-tuning, can derive ego-relevance labels from recorded trajectories rather than a lexicon.

IX. LIMITATIONS

The detector-free VLM does not uniformly beat the detector: its recommended precision is 76.7% versus 79.9%, accuracy is slightly lower, and compute is much higher. A tight power budget still favors the detector-centric design. Four limitations bound our claims. (i) *Provenance*: fine-tuning and evaluation both use ROADWork, and all 25 drives span both splits. Leave-one-city-out replay of the joint estimator costs 0.02–0.04 macro-F1 (Table II), but the fine-tuned model was not retrained per city, and a drive-disjoint split would be stronger. (ii) *Negatives*: a benchmark of hard ego-negatives with category counts would stress the identified failure better. (iii) *Portability*: every cross-model number we report is recalibrated per checkpoint, but the recalibration itself is fitted on one calibration split over a finite grid, so the reported configurations are the best we found rather than optima. (iv) *Baseline reproduction*: our port scores 0.470 against a published 0.600; the split explains +0.029 and frame dropping none (−0.018), leaving about +0.119 to implementation differences, so we claim only that the port is the same pipeline measured under our protocol. Finally, the in-domain data cover two daytime U.S. cities, camera replay is open-loop, and sampled decoding varies by roughly 1–3 points per video (which greedy decoding removes); these results establish neither geographic generalization nor safety readiness.

X. CONCLUSION

An INT4 2B VLM can estimate work-zone state at live-camera rates, matching detector event recall under a latency-honest paired comparison. Testing the assumptions behind that result changes the conclusion. Inherited temporal thresholds move INSIDE IoU by 0.567 absolute, so cross-model comparisons must recalibrate; applying the same search to both models moves only the one that inherited foreign constants. Combining both models under a temporal model fitted for the pair gives the best system inside the detector’s alarm budget,

although most of its gain is the counted temporal model itself. It keeps most of that advantage across cities. The residual ego-relevance gap resists prompting in both models and calls for independently supervised trajectory grounding. We release the replay harness, prompts, metrics and baseline port so that both latency-honest comparison and per-model calibration are reproducible.

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